torch.nn.functional.fractional_max_pool2

https://pytorch.org/docs/stable/generated/torch.nn.functional.fractional_max

Description:

Applies 2D fractional max pooling over an input signal composed of several input planes. Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham The max-pooling operation is applied in kH×kWkH \times kWkH×kW regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes. kernel_size – the size of the window to take a max over. Can be a single number kkk (for a square kernel of k×kk \times kk×k) or a tuple (kH, kW) output_size – the target output size of the image of the form oH×oWoH \times oWoH×oW. Can be a tuple (oH, oW) or a single number oHoHoH for a square image oH×oHoH \times oHoH×oH

Function Signature

```
torch.nn.functional.fractional_max_pool2d(input,
kernel_size, output_size=None, output_ratio=None,
return_indices=False, _random_samples=None)[source]#

Parameters

Examples::
```

Code Examples

```
>>> input = torch.randn(20, 16, 50, 32)
>>> # pool of square window of size=3, and target output size
13x12
>>> F.fractional_max_pool2d(input, 3, output_size=(13, 12))
>>> # pool of square window and target output size being half of
input image size
>>> F.fractional_max_pool2d(input, 3, output_ratio=(0.5, 0.5))
```

Detailed Documentation

Documentation

Applies 2D fractional max pooling over an input signal composed of several input planes.

Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham

The max-pooling operation is applied in kH×kWkH \times kWkH×kW regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes.

kernel_size – the size of the window to take a max over. Can be a single number kkk (for a square kernel of k×kk \times kk×k) or a tuple (kH, kW)

output_size – the target output size of the image of the form oH×oWoH \times oWoH×oW. Can be a tuple (oH, oW) or a single number oHoHoH for a square image oH×oHoH \times oHoH×oH

output_ratio – If one wants to have an output size as a ratio of the input size, this option can be given. This has to be a number or tuple in the range (0, 1)

return_indices – if True, will return the indices along with the outputs. Useful to pass to max_unpool2d().